Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900 - Columns: 18 - Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

I began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

  - Created **age_group** column by binning customer ages.

  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

I performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

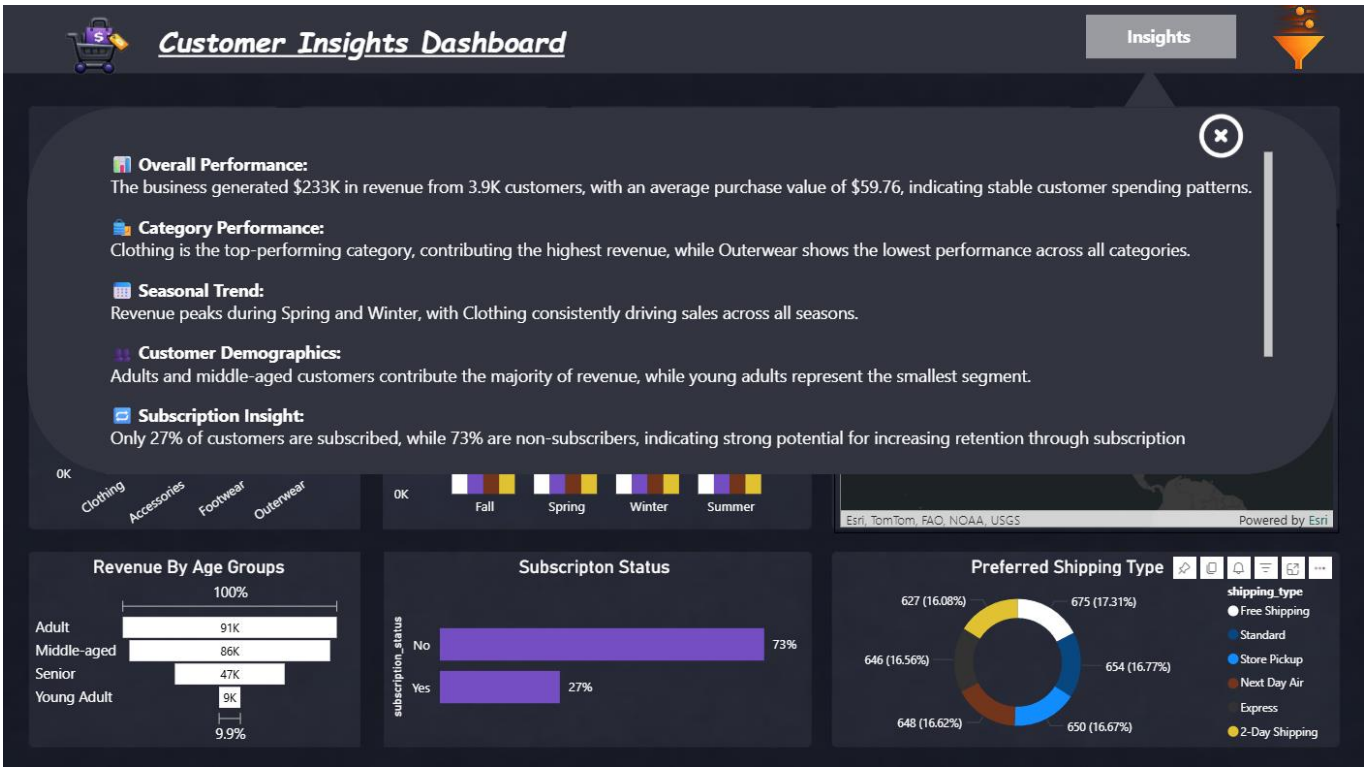
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, I built an interactive dashboard in **Power BI** to present insights visually.



Customer Insights Dashboard
Insights
Filters
category
All
age_group
All
gender
All
item_purchased
All
Purchase Freq
All
location
All
payment_method
All
review_rating
All
season
All
shipping_type
All
REVENUE
$233K
AVG. PURCHASE AMOUNT
$59.76
LOCATIONS
50
AVG. RATINGS
3.75
Revenue by season
category Accessories Clothing Footwear Outerwear
30K
20K
10K
0K
20K 26K 9K 5K
17K 28K 10K 4K
18K 27K 8K 5K
19K 23K 9K 4K
Fall
Spring
Winter
Summer
Esri, TomTom, FAO, NOAA, USGS
Powered by Esri
Adult 91K
Middle-aged 86K
Senior 47K
Young Adult 9K
9.9%
Subscripton Status
subscription_status
No 73%
Yes 27%
Preferred Shipping Type
627 (16.08%)
675 (17.31%)
646 (16.56%)
654 (16.77%)
648 (16.62%)
650 (16.67%)
shipping_type
Free Shipping
Standard
Store Pickup
Next Day Air
Express
2-Day Shipping

Customer Insights Dashboard
Insights
CUSTOMERS
3.9K
REVENUE
$233K
AVG. PURCHASE AMOUNT
$59.76
LOCATIONS
50
AVG. RATINGS
3.75
Top Categories by values
100K
50K
0K
104K
74K
36K
19K
Clothing
Accessories
Footwear
Outerwear
Revenue by season
category Accessories Clothing Footwear Outerwear
30K
20K
10K
0K
20K 26K 9K 5K
17K 28K 10K 4K
18K 27K 8K 5K
19K 23K 9K
Fall
Spring
Winter
Summer
Massachusetts
Location
4K
Total Revenue
72
Total Customers
23
Subscription
25
Total Items
Top Products
Accessories Clothing Footwear
6 6 4 4 4 4 4 4 4
Boots
Dress
Backpack
Coat
Jeans
Sandals
Shirt
Socks
T-shirt
Powered by Esri
Revenue By Age Groups
100%
Adult 91K
Middle-aged 86K
Senior 47K
Young Adult 9K
9.9%
Subscripton Status
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shipping_type
Free Shipping
Standard
Store Pickup
Next Day Air
Express
2-Day Shipping

6. Business Insights

- The business generated $233K in revenue from 3.9K customers, with an average purchase value of $59.76, indicating stable customer spending patterns.
- Clothing is the top-performing category, contributing the highest revenue, while Outerwear shows the lowest performance across all categories.
- Revenue peaks during Spring and Winter, with Clothing consistently driving sales across all seasons.
- Adults and middle-aged customers contribute the majority of revenue, while young adults represent the smallest segment.
- Only 27% of customers are subscribed, while 73% are non-subscribers, indicating strong potential for increasing retention through subscription programs.
- Customer preferences are evenly distributed across shipping methods, with no single dominant delivery option.
- States such as Montana, California, and Idaho show the highest customer activity, highlighting key regions driving engagement.

7. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.